

Flashcards

Q: What are variables in Python?

A: Containers for storing data values (e.g., strings, integers, booleans).

Q: Give examples of Python Data Types

A: Strings, Integers, Booleans, Lists, Tuples, Dictionaries

Q: What are Lists in Python?

A: Ordered, mutable (changeable) collections of items.

Q: What are Tuples in Python?

A: Ordered, immutable (unchangeable) collections of items.

Q: What are Dictionaries in Python?

A: Collections of key-value pairs.

Q: What are Functions in Python and why are they used?

A: Reusable blocks of code that perform specific tasks; promote code reusability and organization.

Q: Explain the purpose of 'if' statements in Python.

A: Allow programs to make decisions based on conditions. ``if condition: execute this``.

Q: Explain 'elif' and 'else' in the context of 'if' statements

A: Provide alternate execution paths when the initial 'if' condition is false.

Q: What are 'for' loops used for in Python?

A: Iterating over collections (strings, lists, ranges).

Q: What are 'while' loops used for in Python?

A: Repeating code as long as a condition is true.

Q: What are Classes in Python?

A: Blueprints for creating objects, encapsulating data and behavior.

Q: What are Objects in Python?

A: Instances of classes.

Q: What is Inheritance in Python?

A: A mechanism where a class inherits attributes and methods from another class, avoiding code duplication.

Q: What are Modules in Python?

A: Provide access to external code and libraries (functions, variables). Use ``import X``.

Q: What is PIP in Python?

A: A package manager for installing and managing third-party Python modules. Use ``pip install X``.